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ORIGINAL PAPER

Key factors determining the seasonal population growth rate of European wild rabbits and their implications for management

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Abstract Recently, the European rabbit (*Oryctolagus cuniculus*) has received contrasting considerations due to its multidimensional role in the Mediterranean ecosystems. Within this framework, knowledge of factors determining its population size may have important consequences for designing an effective management plan. In this paper, we quantified the combined influence of the major demographic and mechanistic factors on seasonal population fluctuation of European rabbits on Lemnos Island (Greece), during 2007–2009. We developed a hypothetical model taking into account direct (productivity, predation, hunting pressure, food shortage, habitat treatment) and indirect factors (soil moisture, adverse weather conditions) using structural equation modeling. We tested for their influence on the seasonal population growth rate (spgr) as determined by line transects to estimate rabbit abundance. The productivity induced higher pgr. Food shortage, which is affected by low soil moisture during the late summer and early autumn, is demonstrated to be the most important negative factor followed by the hunting pressure and predation. Demographic and mechanistic factors highlighted in this analysis could be used either for conservation or for controlling populations of the species.

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Introduction

The European wild rabbit (*Oryctolagus cuniculus*) plays a multidimensional role in Mediterranean type ecosystems (Thompson and King 1994; Devillard et al. 2008). It acts as a disperser of seeds and a primary consumer of plants and seeds and so modifies the native vegetation (see Delibes-Mateos et al. 2008). It is also a digger of soils thereby improving their physicochemical conditions (Delibes-Mateos et al. 2008; but see Eldridge and Simpson 2002; Eldridge et al. 2006) and is prey for many mammalian and avian predators (Serrano 2000; Ferreras et al. 2011). Finally, it is an important game species (Angulo and Villafuerte 2003; Delibes-Mateos et al. 2011). On the contrary, in several areas where it has been introduced, it remains one of the most important threats for biodiversity and human economy (see Thompson and King 1994; Manchester and Bullock 2000), competing with ecologically similar livestock and wild species, destroying both native and cultivated vegetation. As a result, efforts are commonly made to reduce its negative impacts (Myers et al. 1994; Thompson and King 1994; Williams et al. 1995; Courchamp et al. 2003). In the Iberian Peninsula where it is a native species, its population has declined dramatically due to the release of myxomatosis then the subsequent spread of rabbit hemorrhagic disease (RHD; Moreno et al. 2007), causing a number of reactions in ecosystem functions and processes (see Delibes-Mateos et al. 2008; Lees and Bell 2008; but see Barrio et al. 2010b). By contrast, in the south-eastern

Mediterranean basin, it has been introduced on a few Greek islands in the Aegean Sea, but there is little published information concerning its ecology and its impacts on natural ecosystems, or whether it is a pest species or beneficial for the human economy. Rabbits are the main prey for raptorial species (e.g., the Bonelli's eagle *Aquila fasciata* and the common buzzard *Buteo buteo*), and favored by hunters, while at the same time causing extensive damage to agricultural crops having a devastating impact on the local economy during the last two decades (Kontsiotis 2011). Given the contradictory views of the European rabbit in its geographic range and its key role in the Mediterranean basin, its strategic management should be based on an understanding of its ecology; in particular the limiting factors which potentially affect its population growth rate.

The estimation of population growth rate (r or pgr) of a target species on certain time scales has been considered the central issue in population dynamics (see Sibly et al. 2003; Sinclair et al. 2006). Consequently, the determinants of pgr are important factors in understanding their influence on the target species population. In this sense, a wildlife manager should understand all those factors in order to manipulate, increase, maintain, or decrease the pgr depending on the status of the species: endangered, game, or pest, respectively (see Krebs 2001). Factors which influence year-to-year pgr may be demographic (fecundity and mortality), mechanistic (food, parasites, predators, etc.), and density-dependent (see Sibly et al. 2003; Korpimäki et al. 2004), and they act in an unpredictable way on the population. In seasonal environments, like those in the Mediterranean region, year to year population fluctuation is rather negligible (Béltran 1991; Williams et al. 2007).

Although year-to-year pgr of lagomorphs has been studied in detail [see Krebs (2011) for hares; Villafuerte et al. (1997) and Palomares (2003) for wild rabbit], seasonal changes of the European wild rabbit pgr and its determinants are limited in literature (Caley and Morley 2002; Cabrera-Rodriguez 2008). Investigations into the relationships between the European rabbit's seasonal population growth rate (spgr hereafter) and its determinants could lead to a better understanding of its seasonal population fluctuation, as well as assisting integrated management to reduce its abundance in a reasonable population level at the appropriate time (Smith and Trout 1994) or to moderate the extensive damage caused to agricultural crops by population control (Delibes-Mateos et al. 2011).

Consequently, the scope of the present study was to define the demographic and mechanistic factors related to spgr of the European rabbit with the aim of evaluating the combined effects of demographic and mechanistic parameters on the wild rabbit's spgr by using a structural equation model. Implications for managing rabbits are then extensively discussed.

Materials and methods

Study area

The study was conducted in an agricultural area, located in the central region of Lemnos Island (39° 55′ N, 25° 12′ E), in the north Aegean Sea, Greece. It is a lowland area between 8 and 30 m above sea level, dominated by non-irrigated crops (annual cereals, >80 %), with irrigated alfalfa *Medicago sativa*, native vegetation in fallow fields and shrubby riparian vegetation (mixed *Rubus* spp. and *Arundo donax*) occupying the remaining study area. The vegetation in fallow fields is mainly composed of annual winter grasses, legumes, and forbs. The climate is typical-Mediterranean (Csa), with very warm and dry summers and mild winters. The mean annual temperature is 15.9 °C and the average annual precipitation is 474.4 mm, concentrated mainly between November and January. The area experiences occasional thunder storms and heavy rain during June. Mammalian predators are absent in the area. However, the most important predator is the common buzzard (*B. buteo*). The study area is grazed by sheep between June and September, after the cereal crop harvest. The Greek law allows night-shooting from vehicles from October to early March. Unfortunately, the area is also subject to illegal shooting outside this period.

Data collection and parameter determination

A direct method was employed in the current study for assessing rabbits' abundance. Direct methods provide a tool for rapid and accurate estimate of density (Williams et al. 1995), although the detectability of animals and their behavior can undermine their suitability (Barrio et al. 2010a; Fernandez-de-Simon et al. 2011). According to Barrio et al. (2010a), however, direct methods are suitable for the estimation of relative abundance or population trends of wild rabbits in open agricultural landscapes. In addition and in order to minimize the drawbacks of the method, we only surveyed open habitats and only during sunrise and dusk to ensure better sampling of active rabbits (Moreno et al. 2007).

Data were collected during the years 2007–2009. A line transects survey technique was used to estimate the relative density of rabbits. Three fixed line transects were established in the study area and surveyed on consecutive days. Transects were distributed among three neighboring sites which were separated by streams and water channels. This spatial arrangement prevented the movement of rabbits between sites, even though they were spaced at a minimum of 400 m apart. The total length of transects was 8 km (3, 2.5, and 2.5 km each) and the width was 60 m (30 m on either side of the route). Because of the open nature of the habitat it is assumed that within this strip, most rabbits were counted. Surveys were conducted by the same observer with

a mean walking speed of 1.5–2.0 $km h^{-1}$. They were performed only during fine weather and over two daily periods to better sample active rabbits. The first transect counts started early in the morning and finished after 1 h, subsequent counts began in the late afternoon and concluded soon after sunset. Surveys were repeated on each transect at one and a half a monthly intervals (twice a season); early April and mid May in spring, early July and mid August in summer, late September and mid November in autumn, and early January and mid February in winter. In total, 72 replications were carried out during the study period (3 transects × 8 intervals per year × 3 years). Rabbit relative density (individuals ha^{-1}) was calculated by dividing the average number of rabbits counted by the surveyed area.

The spgr was calculated using the formula $r = \log_e \lambda$, where λ is defined by dividing the density in time $t+1$ by the density in time t (Sibly et al. 2003). The mean productivity for each specific survey period (t, $t+1$) was estimated from post-mortem examination of 180 wild rabbits collected monthly by shooting. It was calculated by multiplying the average number of embryos of pregnant females by the percentage of females in the population that were pregnant. In order to calculate those reproductive parameters for the time period (t, $t+1$), we used the average of monthly measurements of the included months for that period. Given that the gestation period of wild rabbits is 28 days, the pregnancy is detectable after 5–7 days (Brambell 1942), and young rabbits emerge at 21 days (Gibb 1990; Williams et al. 1995), we defined that the spgr had been influenced by the productivity with a time delay of one month.

Food shortage was determined by taking into consideration two elements: the frequency of green vegetation, and the quality of food. That data set was collected for each period. The frequency of green vegetation was estimated using line-point intercepts (see Cook and Stubbendieck 1986) along five 25 m linear transects, and expressed as a percentage (%). A total of 500 points were sampled each period, and points were spaced at 0.25 m apart. The concentration of nitrogen was used to measure the quality of food. During each period, we selected 10 square plots (50× 50 cm) which were randomly distributed throughout the study area, and we clipped the vegetation to ground level. Vegetation was dried at 60 °C for 48 h and the crude protein was estimated using the Kjeldahl method (Bremner 1965). Protein content was estimated as $N \times 6.25$ and we used the average content of the total sampled vegetation. The shortage of food was assessed as an inversely ordinal variable with three values: 0=highest percentage (≥70 %) of green vegetation and qualitative food (crude protein of vegetation≥15), 1= limited percentage (40–70 %) of green vegetation with medium quality of food (crude protein of vegetation between 10–15), and 2=almost no green vegetation (<40 %) with the lowest nutritional quality (crude protein of vegetation <10). We assumed that the spgr lagged behind food shortage by one sampling period, since neither the direct effect of deaths caused by starvation nor the indirect effect through the reduction of productivity appear directly related to the initial limitation of vegetation (Boos et al. 2005; Tablado et al. 2009).

Predation pressure was calculated as the number of predators (common buzzards) counted by the same observer at each visit during sampling. Furthermore, between May and July we added three chicks for each pair counted during the sampling. The mean number of chicks was obtained by checking 25 nests on the whole island. In the model, for each time period (t, $t+1$) we used the value of predation which was measured for time t, as this number of predators affects the density of rabbits in that period. The hunting pressure was estimated from the number of cartridges counted along a 2,720-m permanent route, established on a dirt road where both legal and illegal hunting was being practiced. In order to avoid duplicate counting, cartridges were removed after each sampling. In each period, when sampling had ended, a new empty cartridge was left on the route. Cartridges that were probably missed from the last counts and had a higher degree of oxidation, than those which were left, were not considered at the next sampling. In the model, for each time period (t, $t+1$) we used the number of cartridges counted at time $t+1$, as this number of cartridges affect rabbit density in that period. The maximum and minimum temperature, total precipitation and maximum daily precipitation between consecutive surveys (t, $t+1$) were calculated during the three study years. These variables are of particular importance since they have a direct effect on the survival and reproduction of rabbits (Cooke 1977; Gibb and Fitzgerald 1998; Trout et al. 2000; Palomares 2003; Rödel et al. 2009; Rödel and Dekker 2012), and an indirect affect through food shortage. The above variables were represented in the model through a climatic index, which was obtained after performing a principal component analysis on the original climatic data (Calvete et al. 2004; Williams et al. 2007) that measured between consecutive population surveys. Precipitation variables were found to be strongly related to the scores of the first principal component axis, and thus affecting mostly the climatic index.

In each surveying time, we collected five soil samples from each linear transect, located at distances of approximately 500 m. The soil samples were collected at least one week after a rainfall event and not during a dry period (Rueda et al. 2008). The soil moisture was determined as the difference of the weight of dried samples at 100 °C for 24-h by the net weight of soil. We applied management treatments on rabbit warrens during November 2008. One third of rabbits warrens were ripped randomly in our area using diggers and two-wheel-drive tractors with rippers. The destruction of warrens on the selected sites was performed at

approximately 1 m depth, including a 4-m buffer zone around each site. Warrens were first ripped in one direction and then at right-angles to this to completely destroy the warrens (see Williams et al. 1995). Hence, the habitat treatment was introduced as a binary variable in the model; 0 for periods before treatments, and 1 for periods after warrens had been ripped.

Structural equation modeling

Structural equations modeling (SEM) is a powerful statistical method to address causal (direct and indirect) relationships between population parameters and a number of explanatory variables and their interactions. It is a technique for testing and estimating causal relationships, by using a combination of statistical data and qualitative causal assumptions, and it is typically used to develop a modeling strategy (Mitchell 1992). Initially, a path diagram is formed in which the illustrated relationships arise primarily from knowledge based on literature and field experience. Then, estimates of regression weights are computed and the strength of these relationships is depicted in the path diagram (Meyers et al. 2006). The model consists of a series of equations which transform the graphical form of the initial theoretical model to causal linkages among all studied variables (Marcoulides and Schumacker 1996).

The hypothetical model (Fig. 2a) developed assumes that both demographic (productivity) and mechanistic (food shortage, predation, hunting pressure, and habitat manipulation) parameters have a direct effect on spgr; climate has either a direct or an indirect effect, while soil moisture has only indirect effect. Given that neither myxomatosis nor RHD have been recorded in the study area during the last decade, we assumed that the aforementioned parameters regulate the seasonal population fluctuation of the wild rabbit. We also assumed a number of correlations among the independent variables.

Statistical analysis

We used the maximum likelihood procedure to estimate standardized path coefficients. They express the proportions of variance and the correlations among variables. We used different approaches to statistically assess the model fitness: the chi-square test, the normed fit index (NFI), the goodness-of-fit index (GFI), and the root mean square error of approximation (RMSEA; see Browne and Cudeck 1993; Marcoulides and Schumacker 1996). The process of analysis in a structural equations system focuses on minimizing the discrepancy function. The chi-square was also mentioned as a measure of discrepancy. Lower chi-square values indicate better models adaptation. NFI and GFI range between 0 and 1, with values >0.90 indicating a good fit (Marcoulides and Schumacker 1996; Hair et al. 1998), although values higher than 0.8 were considered acceptable (see Marcoulides and Schumacker 1996). RMSEA index values less than 0.05 indicate a good model fit, between 0.05 and 0.08 indicate moderate adjustment, while higher index values indicate poor adjustment (Browne and Cudeck 1993).

SEM analysis was performed with the AMOS (release 7.0 for Windows) procedure of the SPSS statistical package (release 15.0 for Windows).

Results

Population fluctuation

The general pattern of seasonal population trend of the European rabbit is shown in Fig. 1. European rabbits start breeding within the two last weeks of January, whereas substantial numbers of young rabbits appear during spring (from March to May) and summer (June and July). The population decreases gradually from the beginning of autumn (September) until the end of winter (February).

Model evaluation

Path analysis revealed non-statistically significant differences (χ^2=6.896, df=9, P=0.648), indicating a good fit of the collected data to the null model that assumes independence among variables. GFI and NFI suggested a good adjustment, as both indices were greater than 0.90 (0.91 and 0.92, respectively), and the zero value of RMSEA index confirmed a good fit of the model.

Assessment of factors

The values of standardized path coefficients and the values of zero-ordered correlation coefficients are shown in Fig. 2b. The mean productivity had a significant positive effect on spgr (+0.77). On the other hand, significant negative effects on spgr were obtained for food shortage, hunting pressure and predation, but not for management treatment and climate index. Management treatment indicated a negative and climate index had a positive effect, but not significantly so. Soil moisture negatively affected food shortage. The unexplained variance (R) resulted by the model was 0.88.

Discussion

The results of SEM are generally in accordance with our assumptions, since most of the variables were determinants to the spgr of European rabbits in a typical Mediterranean ecosystem. Both demographic and mechanistic parameters,

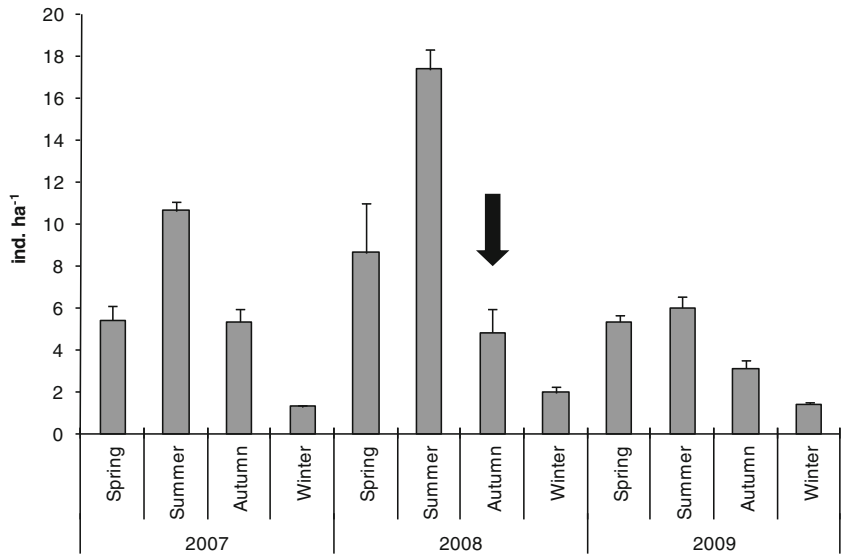

Fig. 1 Population trend (mean seasonal number of individuals per hectare) of European wild rabbit obtained by line transects in Lemnos Island. *Vertical lines above columns* show the standard error (SE) of the mean. The *black arrow* indicates the time that habitat treatment occurred to rabbits' warrens

acting directly on spgr and indirectly through food shortage, explain to a great extent the seasonal population fluctuation of European rabbits. This is in line with the findings reported in the literature (see Williams et al. 1995; Krebs 2001; Caley and Morley 2002; Palomares 2003). However, the unexplained variance indicates that other factors, including density-dependent factors, such as colonization, age structure, parasites, over-winter survival etc. (Thompson and King 1994; Williams et al. 1995; Rödel et al. 2004a, b), are likely to influence the seasonal population variation. Therefore, the model reveals a global picture of the factors causally influencing seasonal population fluctuation, detecting at the same time their combined effect. However, the consideration of several factors simultaneously may partly obscure the significant effect of individual factors, due to their opposing action. For instance, the negative synergistic action of predation and hunting pressure may mitigate the positive effect of productivity on spgr.

Mediterranean ecosystems are highly seasonal environments which tend to confine reproduction and mortality in certain periods during the year. In those environments, a particular fluctuation of population is exhibited, and herbivores, like the European rabbit, take advantage of short favorable seasons to reproduce (di Castri and Mooney 1973). The rapid population growth occurring during spring and early summer is followed by a population decline from early autumn through to the end of winter (Fig. 1). This cyclical fluctuation in population size is apparent and it is further characterized by a high degree of inter-annual variation.

The mean productivity was the main determining factor which is conducive to the European rabbit's spgr. The onset and the end of the breeding season is influenced by environmental and/or ecological factors (e.g., growing vegetation season, precipitation, etc.), and may therefore vary among different areas and among years within an area (Soriguer and Myers 1986; Gonçalves et al. 2002; Rödel and von Holst 2008; Tablado et al. 2009). This variation of the breeding season length may have consequences on the spgr through density fluctuation. Food shortage appears to be the most important negative factor for the spgr. In the Mediterranean region, it could be more pronounced in late summer and early autumn when herbaceous vegetation has dried up and has almost disappeared due to the dry climate (Dallman 1998). Furthermore, the quality of vegetation is degraded drastically during this period, and the available food provided by agricultural crops is minimized due to harvesting (Kontsiotis 2011). All these factors signal the end of the breeding season (Gonçalves et al. 2002; Rödel and von Holst 2008; Tablado et al. 2009) by zeroing the productivity, hence negatively influencing the spgr. In addition, they are responsible for the direct or indirect wild rabbit's mortality, due to starvation or malnutrition (Moreno and Villafuerte 1995; Palomares and Delibes 1997; Villafuerte et al. 1997; Gibb and Fitzgerald 1998; Palomares 2001; Wilson et al. 2002; Williams et al. 2007), by exerting additional negative effects on the spgr, especially during this period where wild rabbit numbers are at their peak (Kontsiotis 2011). Livestock grazing did not have an impact on the available vegetation, since its effect was very low (<1 %) on the total available vegetation for wild rabbits (Kontsiotis 2011).

The negative effect of hunting pressure on wild rabbits has been noted in other studies (Caruso and Siracusa 2001; Angulo and Villafuerte 2003; Williams et al. 2007), but this effect may not be of critical importance in their population dynamics (Williams et al. 1995). In our study, the significant negative effect of hunting pressure could be attributed to its high and constant occurrence throughout the year (both legal and illegal shooting), to the absence of periodically emerging viral diseases (myxomatosis and RHD), and to the low impact of other mortality factors such as human intervention, adverse weather conditions etc.

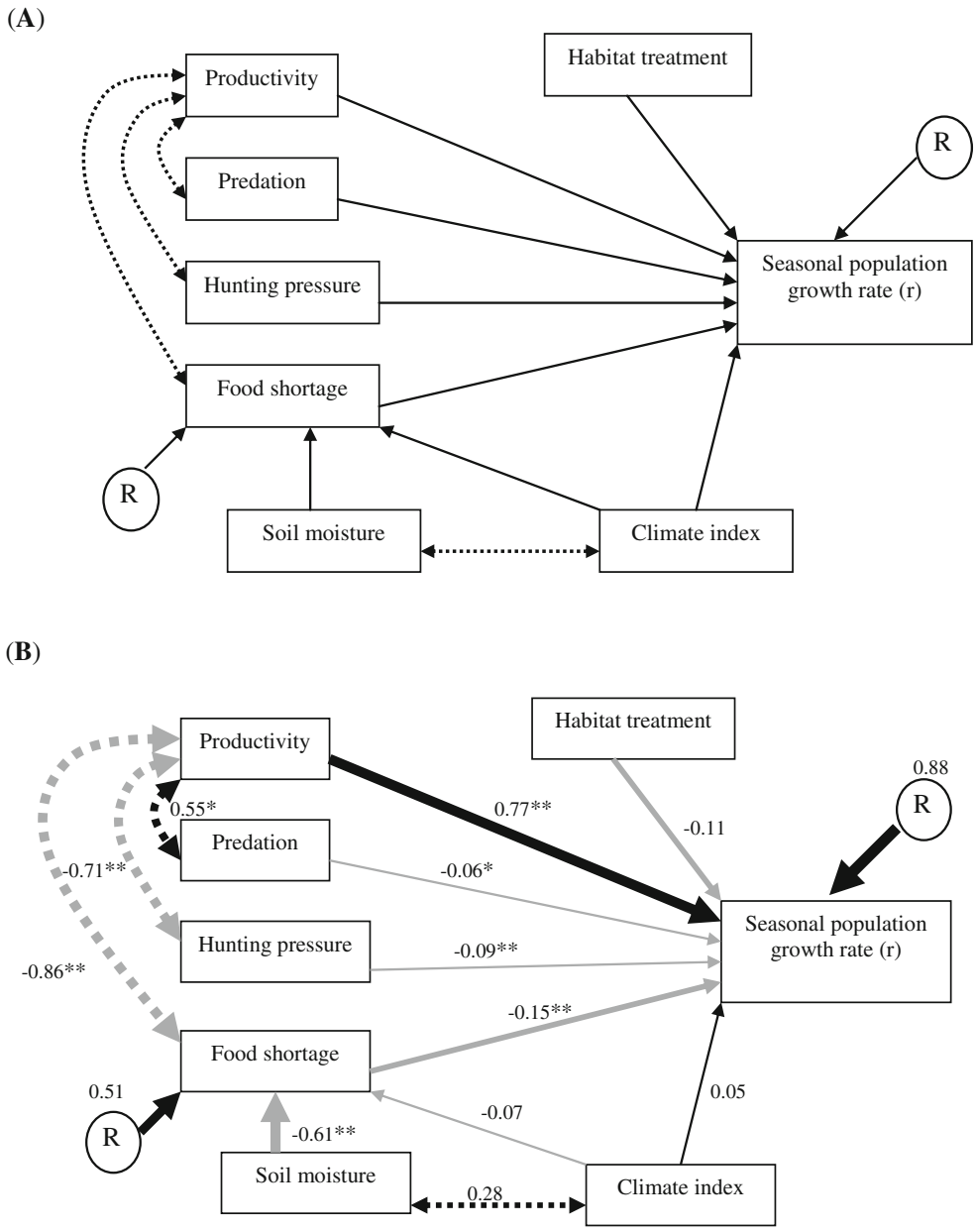

Fig. 2 **a** Structural Equation Model (SEM) depicting hypothetical relationships between both demographic and mechanistic factors, and seasonal population growth rate of European rabbits. *Single-headed arrows* (*solid lines*) indicate causal relationships (standardized coefficients) between variables and *double-headed arrows* (*dashed lines*) show associations (zero-ordered coefficients) between variables. **b** SEM illustrating the direct and indirect effects of demographic and mechanistic factors on the seasonal population growth rate of European rabbits, using density data from line transects (mean relative density). *Black arrows* represent positive effects and *grey arrows* represent negative effects. Arrow widths are proportional to magnitude of coefficients. Values of standardized partial regression coefficients are shown with *asterisks* (*=P<0.05; **=P<0.01) when they are significantly different from zero. *Number above R* denotes the unexplained variance by the model

Predation was a significant parameter influencing spgr of European rabbits in our study. This may be caused by the high breeding density of common buzzards on the island. Several studies have addressed the effect of predation on European rabbits (Moriarty et al. 2000; Lombardi et al. 2003), but the results are rather inconclusive. In the Iberian Peninsula where rabbits constitute the prey for more than 40 predators, the predation displayed a reciprocal negative effect on the predator–prey system (Moreno et al. 2007; Delibes-Mateos et al. 2009), as in most areas of Australia (Gibb and Fitzgerald 1998; Moriarty et al. 2000). Predation is also referred as an important determinant of wild rabbits at low densities, mainly after populations collapse due to diseases (e.g., RHD, myxomatosis), playing an inhibitory role in the population resurgence (Moreno et al. 2007). In England and Wales, Trout et al. (2000) have noted that predator removal was associated with higher numbers of rabbits, but the cause and effect was unclear. In contrast, predation pressure of both red foxes and common buzzards on rabbits was not found by Caruso and Siracusa (2001) in Italy. Furthermore, a relationship between red fox abundance and rabbit abundance was not observed in northeastern Spain (Williams et al. 2007).

Neither habitat treatment nor climate index had significant effects on spgr. Although habitat treatment showed the expected negative effect on spgr, however, the obtained weight was weak and insignificant (Fig. 2b). There are probably two reasons behind this pattern. Firstly, the ripping of the one third of the study area only was apparently not adequate to cause a significant impact. Secondly, although

ripping was suggested by Williams et al. (1995) as the most efficient control measure, especially in agricultural areas (Barrio et al. 2011), it was also suggested that it needs to be combined with additional population control measures, which was not the case in the current study.

Climate index, which in the current study is primarily a function of precipitation, also had an insignificant effect on spgr (Fig. 2b). This was probably due to the good soil drainage, which prevents warren flooding, and the mild climate of the region. It was further found to have a negative but insignificant effect on food shortage, which is explained by the positive effect of precipitation on biomass production. Several researchers have demonstrated the negative impact of adverse weather conditions on European rabbits in areas with prolonged snow cover (Trout et al. 2000), extremely high temperatures (Cooke 1977), and floods (Gibb and Fitzgerald 1998; Palomares 2003; Rödel et al. 2009), as well as interactions between seasonal temperatures and precipitation (Rödel and Dekker 2012).

Significant correlations in zero-ordered coefficients between pairs of variables (e.g., between productivity and predation, see Fig. 2b) are mainly due to their temporal values coincidence. For example, the negative correlation between productivity and food shortage is the result of the opposite temporal appearance of their values; as productivity peaks in spring, food shortage diminishes simultaneously. On the other hand, productivity and predation coincidently both peak during the spring.

In conclusion, the integrative approach made in this study, revealed the importance of demographic and mechanistic parameters to the spgr of the European rabbit. It is further highlighted that the seasonal variability in population density, expressed here as spgr, is determined by direct and indirect interactions between spgr and productivity, food shortage, predation, hunting pressure and soil moisture. Both fecundity and mortality factors regulate the European rabbit's population interannually in Mediterranean ecosystems and generally have considerable potential for its spgr. Therefore, factors determining spgr have profound implications for managing European rabbits (Sinclair et al. 2006), in particular when seasonal population size exceeds a desired level. The effect of warren ripping, as identified in the current study, indicates the need for its application on a larger scale, since its effect on population control is unambiguous.

Implications for management

The management of the European rabbit population is an immediate priority in many regions of the world (Delibes-Mateos et al. 2011), either for control or for conservation purposes (Lees and Bell 2008). Our findings may contribute to a combination of actions and regulations aiming for a multipurpose management of rabbits. For example, a reduction of productivity was motivated by management measures, i.e., the exclusion of rabbits from productive habitats, a simultaneous increase in the hunting intensity and/or an adjustment of hunting period (Gonçalves et al. 2002), could cause a reduction in their populations in areas and/or periods of time where the species is considered a pest. On the other hand, where and when the reduction of rabbits is associated with cascade effects in the ecosystems, stimulation of productivity, e.g., by qualitative food supply, decreasing hunting intensity and/or an adjustment of hunting period (Ferreira and Alves 2009; Gonçalves et al. 2002), could enhance their populations. Even though our study area represents a typical insular Mediterranean agricultural landscape, and a well-defined intra-annually population fluctuation pattern was identified during a three-year study, the extrapolation of our findings to broader scales might be risky and needs to be done with caution (Schaub et al. 2011). Our results suggest that understanding how spgr varies according to both demographic and mechanistic factors is crucial to managing effectively a seasonally fluctuated future rabbit population.

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